

Company Risk Data Collection: Method and Sources

22 September 2026 · Phurinont Klomchitcharoen

Overview

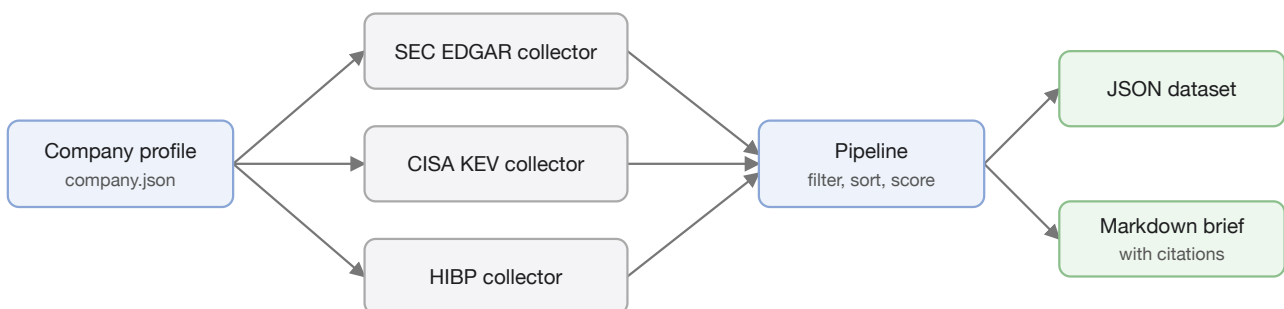
Financial, fraud and cybersecurity risk data for 10 companies was collected on 22 Sep 2026 from three public, keyless sources: SEC EDGAR, the CISA Known Exploited Vulnerabilities catalog and Have I Been Pwned. Every finding cites the source it came from, and every source that was checked is recorded, including the ones that returned nothing.

Companies covered: Microsoft, SAP, Cisco, Salesforce, AWS (collected as Amazon.com, Inc.), Accenture, IBM, GitLab, Chain IQ and HireRight. "Infrastructure" from the original list was not collected because it does not name a single company.

The framework's other two sources were left out of this run: GDELT news (excluded on request) and the sanctions lists (not in scope). Reputational and sanctions categories are therefore not covered.

How the data is collected

Each company is described by a profile, run through three collectors, and written out as a JSON dataset plus a Markdown brief in its own folder.



The collectors run one after another; the pipeline keeps only financial, fraud and cybersecurity findings and scores each category.

- Company profile.** `companies/<name>/company.json` holds the name, aliases, stock ticker, web domain and country used for matching. For AWS the ticker is AMZN, because AWS files no reports of its own.
- Collection.** The command used for each company was `python3 -m risk_collector --company-file companies/<name>/company.json --categories financial fraud cybersecurity --collectors sec_edgar cisa_kev hibp --out companies/<name>`.
- Matching.** Names are normalised (upper-case, accents, punctuation and legal forms such as Inc, Corp, AG and SE removed) and compared with a fuzzy similarity score. SEC companies are found by ticker first, then by name (similarity at least 0.93).
- Sourcing.** A finding (`RiskSignal`) cannot be created without at least one source, and each source needs a name, publisher and URL. Each source records when it was retrieved and, where known, its publication date and licence.
- Negative results.** Every source consulted is logged as a source check with status `ok`, `no_match` or `error`, so "nothing found" is recorded too.

6. **Scoring.** Each category gets a 0–100 triage score: the sum of severity weights (info 0, low 2, medium 6, high 15, critical 40) times confidence, capped at 100.
7. **Output.** Each folder receives `<name>_<timestamp>.json` (all findings and source checks) and `<name>_<timestamp>.md` (a brief with numbered citations, a References list and a source coverage table).

HTTP requests go through a standard-library client with per-host rate limits, up to 3 retries with exponential backoff, and a disk cache in `.cache/risk_collector`.

Sources

Six public endpoints from four publishers were used. None needs an API key, and all SEC, CISA and NVD data is U.S. Government work in the public domain.

Source	Publisher	Endpoint	Categories	Lookback	What it gives	Cache
EDGAR submissions API	U.S. Securities and Exchange Commission	<code>data.sec.gov/submissions/CIK<cik>.json</code>	Financial, fraud, cybersecurity	365 days	Filing history: 8-K items and late-filing notices	6 h
EDGAR full-text search	U.S. Securities and Exchange Commission	<code>efts.sec.gov/LATEST/search-index</code>	Fraud, financial, cybersecurity	365 days	Filings containing red-flag phrases; also used to find each company's SEC ID (CIK)	6 h (7 days for CIK lookup)
XBRL company facts API	U.S. Securities and Exchange Commission	<code>data.sec.gov/api/xbrl/companyfacts/CIK<cik>.json</code>	Financial	Latest two fiscal years	Annual assets, liabilities, equity, current assets and liabilities, net income, revenue, operating cash flow	24 h
CISA Known Exploited Vulnerabilities catalog	U.S. Cybersecurity and Infrastructure Security Agency	<code>cisa.gov/v.../known_exploited_vulnerabilities.json</code>	Cybersecurity	365 days (by date added)	Vulnerabilities in the company's own products that are exploited in the wild	12 h
NIST National Vulnerability Database	U.S. National Institute of Standards and Technology	<code>nvd.nist.gov/vuln/detail/<CVE></code>	Cybersecurity	—	Cited as the detail page for each CVE found in the CISA catalog (not queried)	—
Have I Been Pwned breach catalogue	Have I Been Pwned (Troy Hunt), CC BY 4.0	<code>haveibeenpwned.com/api/v3/breaches</code>	Cybersecurity	5 years (by breach date)	Known data breaches, matched by web domain or name	24 h

SEC's [fair-access policy](#) allows at most 10 requests per second and asks for a User-Agent with contact details, set through `RISK_COLLECTOR_USER_AGENT`. This run used the tool's default User-Agent.

Original filings are cited at `sec.gov/Archives/edgar/data/<cik>/<accession>/<document>`, and HIBP breaches cite the breached organisation's own disclosure page where HIBP lists one.

What counts as a finding

Each source turns raw data into findings with fixed rules, a severity and a confidence (0–1, how sure we are it is about the right company).

SEC filings (confidence 0.95)

Filing	Category	Severity
8-K Item 1.05: material cybersecurity incident	Cybersecurity	Critical
8-K Item 1.03: bankruptcy or receivership	Financial	Critical
8-K Item 2.04: event accelerating a financial obligation	Financial	High
8-K Item 3.01: delisting notice	Financial	High
8-K Item 4.02: non-reliance on past financial statements	Fraud	High
8-K Item 2.06: material impairment	Financial	Medium
8-K Item 4.01: change of auditor	Fraud	Medium
NT 10-K, NT 10-Q, NT 20-F: late annual or quarterly report	Financial	Medium
8-K Item 2.05: restructuring or exit costs	Financial	Low

Filing text searches (confidence 0.5)

Phrase searched	Category	Severity
"Wells notice"	Fraud	Medium
"deferred prosecution agreement"	Fraud	Medium
"substantial doubt" + "going concern"	Financial	Medium
"material weakness"	Fraud	Low
"ransomware"	Cybersecurity	Low
"cybersecurity incident"	Cybersecurity	Low

These are kept low because the phrases often appear in standard risk-factor wording. Each finding cites up to 5 matching filings.

Financial ratios from XBRL (confidence 0.9)

Rule	Severity
Negative shareholders' equity	High
Net loss two years in a row	High (one year: Medium)
Revenue down more than 25% year over year	High (more than 10%: Medium)
Current ratio below 0.7	High (below 1.0: Medium)
Liabilities above 90% of assets (not applied to financial firms)	Medium
Negative operating cash flow (not applied to financial firms)	Medium
Annual snapshot of all metrics	Info

Cybersecurity feeds (confidence 0.9)

- **CISA KEY:** a vulnerability matches when its vendor name is at least 0.92 similar to the company name or an alias. Severity is High if CISA knows it is used in ransomware campaigns, otherwise Medium. At most 25 per company are listed.
- **HIBP:** a breach matches on the company's web domain (confidence 0.9) or on its name at 0.95 similarity (confidence 0.6). Severity is High if 1 million or more accounts were exposed or the data includes passwords, card or bank numbers, social security numbers or government IDs; otherwise Medium.

Limitations

Findings are signals for an analyst to review, not conclusions, and coverage is uneven across the 10 companies.

- **Private and non-U.S. companies are thinly covered.** Chain IQ is a private Swiss company with no SEC filings, so only the CISA and HIBP checks apply. HireRight was taken private in 2024; its SEC record still exists but has no financial data (the request returned HTTP 404).
- **AWS is measured as Amazon.** Financials and filings are for Amazon.com, Inc. as a whole.
- **Keyword hits can be false positives.** "Material weakness" or "ransomware" in a filing is often standard wording rather than an actual incident.
- **The CISA list is capped at 25 per company.** Microsoft had 48 matches in the last year; the brief lists 25 and the source check records the full count.
- **Old vulnerabilities can appear.** The 365-day window uses the date CISA added a vulnerability, so an old flaw such as CVE-2008-4128 appears if it was added recently.
- **Scores are for triage only.** They are a simple weighted sum, not a calibrated rating.
- **Not covered in this run:** news (GDELT), sanctions lists, and any regulator or court databases outside the U.S. SEC.